

Exposé IX, continuation. Néron models and monodromy

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3. The semi-stable reduction case. The theorem of semi-stable reduction

We first record an auxiliary result in the style of SGA 3.

Proposition 3.1. Let S be a scheme, let G be a commutative S -group scheme, flat, separated, and of finite presentation over S , let H be a quasi-separated S -group scheme locally of finite type, and let

$$u : G \longrightarrow H$$

be a homomorphism of S -group schemes. Put $K = \text{Ker } u$. For $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $k(s)$, and write $\mu(s)$, respectively $\alpha(s)$, for the reductive rank, respectively the abelian rank, of $K_{\bar{s}}$, i.e. the dimension of the largest subtorus, respectively of the largest abelian variety quotient, of $(K_{\bar{s}})_{\text{red}}^0$. Let $s \in S$, let $t \in S$ be a generalization of s , and let p be the characteristic exponent of $k(s)$.

- a) For every integer $n > 0$ prime to p , the group ${}_nK$ is, in a neighbourhood of s , finite, separated and étale over S , and

$$\text{rank}({}_nK_s) \mid \text{rank}({}_nK_t). \quad (3.1.1)$$

- b) One has

$$\mu(s) + 2\alpha(s) \leq \mu(t) + 2\alpha(t). \quad (3.1.2)$$

- c) If $(K_t)_{\text{red}}^0$ is unipotent, then so is $(K_s)_{\text{red}}^0$.
- d) Suppose G_s has unipotent rank zero, that H is smooth over S , and that u_t is an isogeny. Then u is flat and quasi-finite over an open neighbourhood U of s ; in particular $K|_U$ is flat and quasi-finite.
- e) Under the hypotheses of d), if u_t is moreover a monomorphism, hence an isomorphism of G_t onto an open subgroup of H_t , then u_s is an isomorphism of G_s onto an open subgroup of H_s . If moreover S is irreducible with generic point t , then, after replacing S by a neighbourhood of s , the morphism u induces an isomorphism of $G|_U$ onto an open subgroup of $H|_U$.

The proof is by reduction to standard facts on finite flat group schemes and on the openness of the unit section of a separated formally étale group. For a), one may suppose that n is invertible on S . Then multiplication by n on G is étale fibre by fibre, hence étale; the kernel ${}_nG$ is finite over S , and the kernel ${}_nK$ is an open subgroup of ${}_nG$. After strict henselization at s , ${}_nK$ decomposes into a finite part and a part with empty special fibre, giving the divisibility (3.1.1), and even an injection on geometric fibres

$${}_nK(\bar{s}) \hookrightarrow {}_nK(\bar{t}).$$

Raising the two sides to powers in n gives (3.1.2), since the torsion of a commutative algebraic group detects the sum $\mu + 2\alpha$ away from p . Assertion c) follows formally from b). For d), the generic isogeny implies that the generic kernel is finite; the hypotheses on dimensions and on the unipotent rank of G_s force the special kernel to be finite, and the SGA 3 flatness criterion gives flatness and quasi-finiteness in a neighbourhood. Assertion e) follows from d) and the open immersion criterion.

Lemma 3.1.3. Let K be quasi-finite, of finite presentation, separated and flat over S . If $s \in S$ and t is a generalization of s , then

$$\text{rank } K_s \leq \text{rank } K_t.$$

If $\text{rank } K_\eta \leq 1$ for every maximal point η of S , then $K \rightarrow S$ is an open immersion; hence it is an isomorphism when it has a section. The proof is local on S . In the strictly local case one separates off the finite open-and-closed part with non-empty special fibre; flatness and finite presentation make its rank constant.

Remark 3.1.4. In 3.1 c) and in the sequel to Proposition 3.1, no information is supplied about ${}_n K_s$ when n is a power of p . If G is smooth, or at least if the relevant finite-presentation hypotheses are imposed on H , the same method gives an analogous statement for the multiplicative rank. Namely

$$\text{rank}_{\text{mult}}({}_n K_s) \mid \text{rank}_{\text{mult}}({}_n K_t), \quad (3.1.4.1)$$

where $\text{rank}_{\text{mult}}$ denotes the rank of the largest subgroup of multiplicative type. A torsion group of multiplicative type is finite, so this again implies the unipotence assertion.

Proposition 3.2. Let S be a locally noetherian regular integral scheme of dimension 1, let K be its function field, let A_K be an abelian scheme over K , and let A be its Néron model over S . Thus A is a smooth separated group scheme over S , and it is quasi-projective when S is noetherian. The following conditions are equivalent:

- (i) for every $s \in S$, A_s^0 has unipotent rank zero; equivalently A_s^0 is an extension of an abelian scheme by a torus;
- (ii) there is a smooth separated finite-type group scheme G over S , extending A_K , whose fibres have unipotent rank zero.

Moreover, if G is as in (ii), the unique morphism $G \rightarrow A$ inducing the identity on generic fibres is an isomorphism of G onto an open subgroup of A , and therefore induces an isomorphism on neutral components,

$$G^0 \xrightarrow{\sim} A^0.$$

The definition of the Néron model here is the same as in the trait case recalled earlier; its existence and its separatedness of finite type are local on the base. The implication (i) $\Rightarrow$ (ii) is obtained by taking $G = A$. Conversely, if G is as in (ii), Proposition 3.1 e) says that $G \rightarrow A$ is an open immersion.

Corollary 3.3. With the notation of 3.2, suppose that the neutral components of the fibres of the Néron model have unipotent rank zero. If S' satisfies the same hypotheses as S , if K' is its function field, and if $f : S' \rightarrow S$ is dominant, then the canonical morphism from the pullback of A to the Néron model A' of $A_K \otimes_K K'$ is an open immersion and induces an isomorphism

$$A^0 \times_S S' \xrightarrow{\sim} A'^0. \quad (3.3.1)$$

This follows from 3.2 applied to S' , taking $G = A \times_S S'$.

Definition 3.4. In the notation of 3.2, the abelian scheme A_K over K is said to have *semi-stable reduction over S* when the equivalent conditions (i) and (ii) of Proposition 3.2 hold. Equivalently, this may be tested after localizing at traits, and even after passing to the henselization or strict henselization.

Proposition 3.5 (Galois criterion for semi-stable reduction). Assume that S is a trait. Keep the notation of 2.5.3 after passing to the henselization of S . The following conditions on A_K are equivalent:

(i) A_K has semi-stable reduction over S ;

(ii) with the notation of (2.5.3) and (2.5.4),

$$V \subset V^\perp;$$

(iii) for the toric part W of the Tate module,

$$W = V^\perp;$$

(iv) the inertia group I acts on $U = T_\ell(A(\bar{K}))$ unipotently of echelon 2: there exists an I -stable submodule $U' \subset U$ such that I acts trivially on U' and on U/U' .

Equivalently, if g is a topological generator of $\mathbb{Z}_\ell(1) = T_\ell(\bar{K}^*)$, the action of inertia factors through the maximal pro- ℓ finite quotient and

$$(1 - g_U)^2 = 0. \tag{3.5.3}$$

The proof uses the orthogonality theorem of Section 2: condition (ii) says $\lambda = 0$, hence is equivalent to the semi-stability of the neutral component; (ii) and (iii) are equivalent by the orthogonality $W = V \cap V^\perp$, and (iv) is the same assertion expressed through the dual action of inertia.

Theorem 3.6 (semi-stable reduction theorem). Let S be a connected regular noetherian scheme of dimension 1, let K be its function field, and let A_K be an abelian scheme over K . Then there is a finite Galois extension K'/K such that $A_K \otimes_K K'$ has semi-stable reduction over the normalization S' of S in K' .

It suffices to prove this after replacing S by traits at the finitely many closed points where A_K does not have good reduction. The Galois group of a finite normal extension permutes the points above a fixed closed point, so semi-stability at one such point suffices. By the criterion 3.5, the assertion is reduced to the existence of an open subgroup of inertia acting unipotently of echelon 2 on $T_\ell(A(\bar{K}))$, which follows from the monodromy theorem already established.

Corollary 3.7. If S is a trait, there exists an open subgroup I' of the inertia group I such that the action of I' on $U = T_\ell(A(\bar{K}))$ is unipotent of echelon 2. This is equivalently the existence of an open subgroup π' of the fundamental group for which the induced inertia action has this property.

Corollary 3.8. The equivalent conditions of Proposition 3.5 are also equivalent to the condition that the action of I on U be unipotent. Indeed, if an open subgroup acts unipotently of a given echelon on a finite-dimensional $\mathbb{Q}_\ell$ -vector space, then the Zariski closures defined by the full group and by the open subgroup coincide, since a unipotent algebraic group in characteristic zero is connected.

Corollary 3.9. In the notation of 3.2, the two conditions of Proposition 3.2 are also equivalent to the following stability property: for every finite étale extension K'/K , if S' is the normalization of S in K' and A' is the Néron model of $A_K \otimes_K K'$, then the canonical morphism

$$A \times_S S' \longrightarrow A'$$

induces an isomorphism on the neutral components of the fibres.